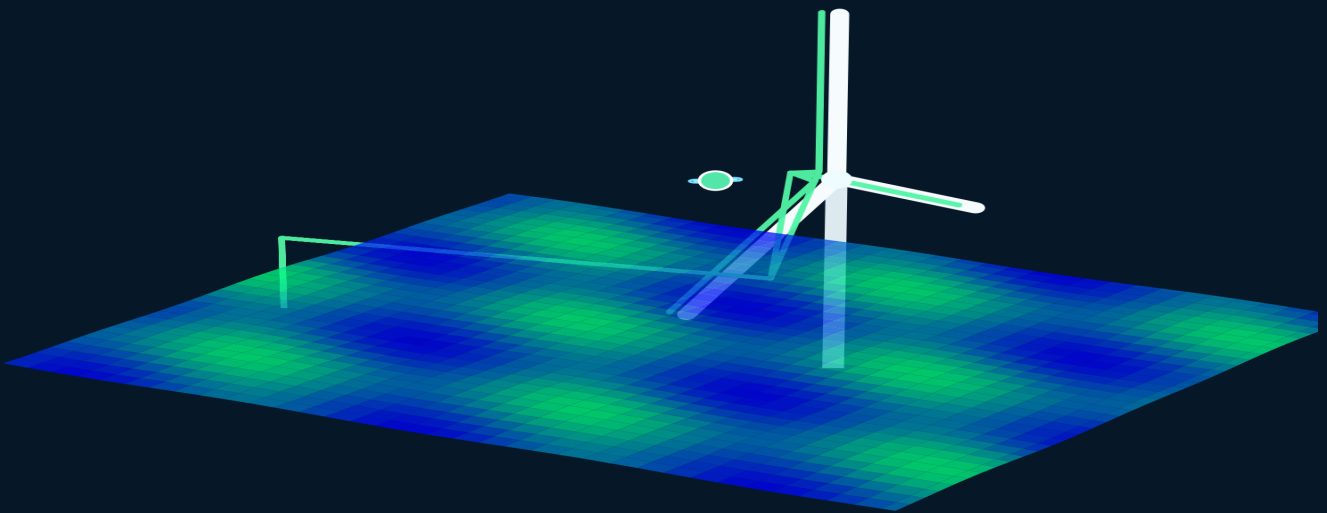


CONCEPT-LEVEL OFFSHORE INSPECTION DIGITAL-TWIN PROTOTYPE

Offshore Autonomous Blade Inspection Mission Simulator V2

Professional User Guide | Version 2.0.0



Purpose

A concept-level systems-engineering simulator for blade-by-blade autonomous offshore wind-turbine inspection, including mission geometry, coverage, blade-relative navigation, energy-aware return logic, environmental effects, safety indicators, economics, traceability, and exportable results.

Important: This application is not validated flight-dynamics, operational-safety, collision-avoidance, or defect-detection software.

What the simulator does

The application evaluates a stationary-rotor inspection concept from launch through landing, then presents an engineering disposition and supporting evidence.

Capability	What the model evaluates
Mission construction	Takeoff, transit, blade transition, two inspection passes per selected blade, hub clearance, egress, return, and landing.
Coverage	A simplified blade-surface grid combined with camera footprint, overlap, LiDAR range, and data-quality gates.
Energy	Induced, profile, parasitic, environmental, and hotel loads, plus predicted return energy and landing reserve.
Navigation	Scenario-based GNSS uncertainty reduced by illustrative optical, LiDAR, IMU, and synchronization fusion credit.
Safety	Power feasibility, energy margin, state of charge, stand-off error, navigation error, and an ordinal rotor-proximity hazard index.
Economics	Scenario-dependent annual generation and revenue-at-risk calculations based on user-entered assumptions.

Core disposition logic

Result	Typical basis
SIMULATED MISSION ACCEPTABLE	Modeled coverage, energy, navigation, and hazard criteria are satisfied.
SIMULATED MISSION CONDITIONAL	The flight completes, but one or more concept-level acceptance criteria are not fully satisfied.
SIMULATED MISSION INFEASIBLE	The model terminates the mission or indicates excessive navigation or proximity risk.

Interpretation rule

Treat all results as scenario-screening evidence. The data-suitability and hazard indices are uncalibrated engineering indicators, not probabilities of defect detection or collision.

Run a mission in four steps

Use the sidebar to define a coherent turbine, environment, UAS, sensor, and economic scenario.

Configure the turbine

- 1
- Enter rating, capacity factor, hub height, rotor diameter, blade-root offset, blade count, rotor state, and rotor speed.

Define the environment

- 2
- Set wind, gust, turbulence, wave height, visibility, precipitation, launch platform, and navigation condition.

Define the aircraft and sensors

- 3
- Enter UAS mass, propulsion and battery values, speeds, stand-off, launch distance, camera, LiDAR, IMU, and synchronization inputs.

Run and review

- 4
- Select **Run autonomous inspection**, then review the disposition, mission prototype, telemetry, coverage, safety, economics, and exports.

Recommended baseline scenario

Group	Suggested starting values
Turbine	15 MW, 48% capacity factor, 135 m hub height, 236 m rotor diameter, 3 blades, parked and secured.
Environment	9 m/s mean wind, 12 m/s gust, 10% turbulence, 1.5 m waves, 10 km visibility, no rain, degraded GNSS.
UAS	14.5 kg, quadrotor, 0.65 m propellers, 3,200 Wh battery, 30% reserve, 8 m/s cruise, 1.2 m/s inspection, 5 m stand-off.
Sensors	70 x 50 degree camera field of view, 2 Hz capture, 82% optical and IMU quality, 60 m LiDAR, 2 cm noise, 20 Hz update, 12 ms synchronization error.

Geometry check

The lower blade-tip clearance must be at least 15 m: hub height minus one-half rotor diameter. Invalid geometry prevents the simulation from running.

Mission configuration reference

Inputs are grouped by engineering function. Values should represent one internally consistent mission concept.

Group	Key inputs	Engineering role
Turbine	Rating, capacity factor, hub height, rotor diameter, root offset, blades, rotor state, rpm	Defines inspection geometry, energy-production context, and rotor-state limitations.
Environment	Wind, gust, turbulence, waves, visibility, precipitation, launch platform, navigation condition	Modifies power, tracking uncertainty, image quality, LiDAR quality, and hazard.
UAS and energy	Mass, rotor count, propeller diameter, figure of merit, efficiency, CdA, max power, battery, SOC, reserve, speeds, stand-off, launch distance	Drives propulsion load, mission duration, return feasibility, and power-limit checks.
Sensors	Camera FOV and rate, optical quality, LiDAR range/noise/rate, IMU quality, synchronization error, overlap	Drives fusion credit, image footprint, overlap, data-quality screening, and coverage.
Economics	Electricity value, assumed AEP loss, turbine count	Estimates scenario-dependent annual revenue exposure.

Critical input cautions

- **Rotor state:** Parked and secured is the supported concept. Rotating-rotor selections are exploratory stress cases because time-dependent blade tracking is not implemented.
- **Mass:** A selected takeoff mass at or above approximately 24.95 kg is flagged as outside the usual U.S. Part 107 small-UAS weight range.
- **Altitude:** The inspection volume can exceed 121.9 m (400 ft) above the local water datum and may require authorization in an actual operation.
- **Economic assumptions:** Capacity factor is annual and is not inferred from mission-day weather. Revenue at risk is not guaranteed UAS savings.

Input discipline

Do not use default values as evidence that a real aircraft, sensor suite, or offshore operation is feasible. Replace defaults with traceable assumptions, vendor data, test data, or documented design targets.

How to read the results

The summary cards show the highest-value mission indicators. Detailed evidence appears in the tabs.

Metric	Meaning	Use
Geometric coverage	Percentage of the simplified blade-surface grid observed under the model's footprint and quality gates.	Assess whether the planned path provides sufficient geometric observation.
Final state of charge	Remaining battery percentage when the active mission path ends.	Compare with the selected landing reserve.
Minimum energy margin	Remaining energy minus predicted return, landing allowance, and reserve.	Negative values trigger mission termination.
P95 stand-off error	95th-percentile deviation from desired blade stand-off during inspection.	Evaluate proximity-control precision.
P95 navigation error	95th-percentile blade-relative navigation uncertainty during inspection.	Compare with the illustrative 0.50 m target.
Hazard index	Ordinal, uncalibrated proximity-risk indicator from environment, navigation, stand-off, and rotor-state factors.	Screen scenarios and identify drivers requiring mitigation.
Data suitability	Composite index using coverage, overlap, navigation, stand-off, LiDAR quality, and energy.	Compare scenarios only. Do not interpret as defect-detection probability.

Tab guide

Tab	Primary content
Mission Prototype	3D turbine, ocean, active flight path, mission phases, and stationary-rotor concept trajectory.
Telemetry	State of charge and hazard history, plus time-step data for power, energy margin, navigation, and stand-off.
Coverage and Data	Radial heatmap of covered blade-grid cells and a clear statement of model limitations.
Safety	Hazard index and scenario-specific recommendations.
Economics	Annual generation and revenue-at-risk estimates.
Systems Engineering	Need-to-requirement-to-verification traceability table.
Report and Export	Markdown report, telemetry CSV, and JSON package.

What happens during a simulation

The simulator builds a discrete mission path, annotates every step with physics and uncertainty, then aggregates the active portion into a mission summary.

Stage	Model behavior
1. Build path	Creates takeoff, transit, blade transition, two inspection passes, hub-clearance, egress, return, and landing segments using smooth interpolation.
2. Assign speed and time	Maps a commanded speed to each phase, computes segment distance, time step, and elapsed time.
3. Estimate navigation	Maps the selected navigation condition to baseline uncertainty, then applies phase-dependent multisensor fusion credit and environmental penalties.
4. Estimate stand-off	Measures geometric distance from the flight point to the nearest blade-line point, then adds deterministic-seed scenario noise.
5. Estimate power and energy	Combines induced, profile, parasitic, environmental, phase, and hotel loads. Energy use is integrated over time.
6. Protect return reserve	At each step, predicts energy needed to return and land. The mission becomes inactive if energy margin, SOC, or power feasibility is violated.
7. Estimate data quality	Computes overlap, LiDAR quality, image footprint, and valid grid-cell observations.
8. Summarize	Calculates mission duration, coverage, P95 errors, hazard, data suitability, economics, disposition, and recommendations.

Reproducibility

The random-number seed is fixed at 42. Identical inputs should produce identical illustrative stochastic results in the same software environment.

Not included

No full six-degree-of-freedom dynamics, aerodynamic wake model, vessel-motion coupling, calibrated collision probability, validated autonomy stack, validated defect detector, or time-dependent rotating-blade interception.

Use the model as a screening tool

The application is best suited to early concept development, trade studies, requirements discussion, and scenario comparison.

Suggested engineering workflow

- **Define the decision:** State what design choice or operational assumption the simulation is intended to inform.
- **Establish traceable inputs:** Record the source, date, uncertainty, and authority for each nontrivial value.
- **Run sensitivity cases:** Vary wind, gust, navigation condition, battery capacity, stand-off, launch distance, inspection speed, and sensor quality.
- **Inspect failure modes:** Determine whether infeasibility is driven by power, energy margin, navigation, proximity, or coverage.
- **Document limitations:** Keep the concept-level disclaimer with every exported result.
- **Escalate evidence:** Use higher-fidelity simulation, HIL testing, controlled flight tests, and formal safety assessment before operational decisions.

Example traceability in the application

Need ID	Requirement ID	Requirement	Verification
SN-01	SYS-1.1	Conduct remote blade inspection without personnel entering the rotor inspection envelope.	ConOps review and demonstration
SN-02	NAV-1.1	Maintain P95 blade-relative position error no greater than 0.50 m under the defined degraded-GNSS test condition.	HIL and controlled flight test
SN-03	INS-1.1	Achieve at least 90% geometric coverage of the defined blade inspection surface.	Simulation and reference-target inspection
SN-04	ENE-1.1	Preserve energy for predicted return, landing, and selected reserve.	Energy analysis and flight-log review

Verification distinction

A favorable simulator result is analysis evidence, not verification closure. Requirements that affect flight safety, navigation performance, inspection effectiveness, or energy reserve require independent verification in representative conditions.

Preserve assumptions and evidence

The application produces three export formats. Keep them together for configuration control.

Export	Contents	Recommended use
Markdown report	Mission ID, timestamp, simulator version, disposition, results, economics, recommendations, and limitations.	Human-readable review package and design-record attachment.
Telemetry CSV	Time-step mission state including phase, position, speed, stand-off, navigation error, power, SOC, return energy, margin, overlap, LiDAR quality, hazard, and activity.	Engineering analysis, plotting, audit, and sensitivity comparison.
Simulation JSON	Complete input set, summary object, model version, and random seed.	Reproducibility, automated comparison, and configuration archival.

Recommended file package

- Use a unique mission ID for every baseline and sensitivity case.
- Store the Markdown report, telemetry CSV, and JSON export in the same controlled folder.
- Record the application commit or release identifier in addition to Version 2.0.0.
- Retain the external assumptions and source documents used to set turbine, vehicle, sensor, environmental, and economic inputs.
- Do not remove the concept-level limitations statement from reports used in academic or design reviews.

Example naming convention

OW-UAS-V2-001_baseline_report.md
OW-UAS-V2-001_baseline_telemetry.csv
OW-UAS-V2-001_baseline_simulation.json

Guide scope

This guide reflects the supplied Version 2.0.0 source code and its visible user interface, calculation flow, outputs, safeguards, and stated limitations. It does not certify the underlying assumptions, software implementation, or operational suitability.